Surds

Number · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10 · worksheet

A+ Maths Hub

Name	Class	Date

Answer ALL questions.

Write your answers in the spaces provided. You must write down all the stages in your working.

Suggested time: 108 minutes · Total marks: 108

KEY RULES

Multiply	$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$
Divide	$\frac{\sqrt{a}}{\sqrt{b}} = \sqrt{\frac{a}{b}}$
Simplify	$\sqrt{50} = \sqrt{25 \times 2} = 5\sqrt{2}$
Rationalise	$\frac{1}{\sqrt{a}} = \frac{\sqrt{a}}{a}$; $\frac{1}{a+\sqrt{b}} \rightarrow$ multiply by $\frac{a-\sqrt{b}}{a-\sqrt{b}}$
Expand	Use FOIL; $(\sqrt{a})^2 = a$

WORKED EXAMPLE 1

[2 marks]

Write $\sqrt{50}$ in the form $a\sqrt{2}$.

Step 1	Write 50 using a square number	$\sqrt{25 \times 2}$
Step 2	Split the root	$\sqrt{25} \times \sqrt{2} = 5\sqrt{2}$

Answer: $5\sqrt{2}$

WORKED EXAMPLE 2

[3 marks]

Expand and simplify $(3+\sqrt{2})(3-\sqrt{2})$

Step 1	Multiply every term by every term	$9 - 3\sqrt{2} + 3\sqrt{2} - 2$
Step 2	The surd terms cancel	$9 - 2$

Answer: 7

WORKED EXAMPLE 3

[3 marks]

Rationalise the denominator of $\frac{6}{\sqrt{3}}$

Step 1	Multiply top and bottom by $\sqrt{3}$	$\frac{6\sqrt{3}}{3}$
Step 2	Simplify	$2\sqrt{3}$

Answer: $2\sqrt{3}$

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1 Write $\sqrt{18}$ in the form $a\sqrt{2}$.

(Total for question = 1 mark)

Practice question written for this worksheet (not from a past paper)

2 Write $\sqrt{98}$ in the form $a\sqrt{2}$.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

3 Simplify $\sqrt{72}$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

4 Simplify $2\sqrt{3} + 5\sqrt{3}$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

5 Simplify $\sqrt{50} + \sqrt{8}$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

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6 Write $5\sqrt{6}$ as a single square root.

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

7 Rationalise the denominator of $\frac{10}{\sqrt{5}}$

(Total for question = 2 marks)

Practice question written for this worksheet (not from a past paper)

8 Rationalise the denominator of $\frac{5+\sqrt{5}}{\sqrt{5}}$

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

9 Expand and simplify $(2+\sqrt{5})^2$

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

10 Write $\frac{6}{3+\sqrt{3}}$ in the form $a-\sqrt{b}$.

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

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11 Simplify $\dfrac{\sqrt{125}+\sqrt{80}}{\sqrt{5}}$

(Total for question = 3 marks)

Practice question written for this worksheet (not from a past paper)

12 **13** $a=\sqrt{8}+4$

$b=\sqrt{8}-4$

$(a-b)(a+b)$ can be written in the form $y\sqrt{4y}$

Find the value of y
Show your working clearly.

$y=$ .....

(Total for Question 13 is 3 marks)

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13 Without using a calculator, show that $\dfrac{12}{\sqrt{2}-1}-\left(\sqrt{2}\right)^5=2\sqrt{32}+12$

Show your working clearly.

(Total for question = 3 marks)

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14 Show that $\frac{4}{3\sqrt{5}+7}$ can be written in the form $a-\sqrt{b}$ where a and b are integers.

Show each stage of your working.

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15 Show that $\frac{2\sqrt{3}}{\sqrt{3}-1}$ can be written in the form $a+\sqrt{a}$ where a is an integer

Show your working clearly.

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16 Show that $\dfrac{4+\sqrt{8}}{\sqrt{2}-1}$ can be written in the form $a+b\sqrt{2}$, where a and b are integers.

Show each stage of your working clearly and give the value of a and the value of b.

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17 Show that $\frac{2\sqrt{7}+2}{\sqrt{7}-3}$ can be written in the form $a-\sqrt{b}$ where a and b are integers.

Show your working clearly.

(Total for question = 3 marks)

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18 Show that $\frac{1+\sqrt{5}}{3-\sqrt{5}}$ can be written in the form $a+\sqrt{b}$ where a and b are integers.

Show each stage of your working clearly.

(Total for question = 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2024 · Question 17

19 Show that $\left(4-\sqrt{20}\right)\left(5-2\sqrt{5}\right)$ can be written in the form $a-\sqrt{b}$

where a and b are integers.

(Total for question = 3 marks)

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20 Show that $\frac{\sqrt{12}}{\sqrt{3}+2}$

can be written in the form $a+\sqrt{b}$ where a and b are integers

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21 Express $\frac{8}{\sqrt{5}-1}$ in the form $\sqrt{a}+b$ where a and b are integers

Show each stage of your working clearly.

Edexcel IGCSE Mathematics A (4MA1) · Paper 2H · November 2021 · Question 17

22 Given that $(8-\sqrt{x})(5+\sqrt{x})=y\sqrt{x}+21$ where x is a prime number and y is an integer,

find the value of x and the value of y.
Show each stage of your working clearly.

x = ...

y = ...

(Total for question = 3 marks)

Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2021 · Question 18

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23 Show that $\frac{\sqrt{8}}{\sqrt{8}-2}$ can be written in the form $n+\sqrt{n}$, where n is an integer.

Show your working clearly.

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · January 2019 · Question 18

24 Without using a calculator, rationalise the denominator of $\frac{6}{3-\sqrt{7}}$

Simplify your answer.
You must show each stage of your working.

.....

(Total for question = 3 marks)

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25 $(\sqrt{3})^5 = k\sqrt{3}$ where k is an integer.

(a) Find the value of k

$k =$ .....

(1)

(b) Show that $\dfrac{21}{3-\sqrt{2}}$ can be written in the form $c + \sqrt{d}$ where c and d are integers.
Show each stage of your working clearly.

(3)

(Total for question = 4 marks)

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26

16 (a) Rationalise the denominator of $\frac{a+\sqrt{4b}}{a-\sqrt{4b}}$ where a is an integer and b is a prime number.

Simplify your answer.

(3)

(b) Given that $\left(\sqrt{\frac{y}{x}}\right)^{-5} = \frac{x^m}{y^m}$ where $x \neq y$

find the value of m.

$m =$

(1)

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27 (a) $\left(\sqrt[4]{k^{12}}\right)^5 = k^n$

Find the value of n

$n =$ .....

(1)

(b) Express $\dfrac{7}{2-\sqrt{3}}$ in the form $\sqrt{c} + d$ where c and d are integers.

Show your working clearly.

.....

(3)

(Total for question = 4 marks)

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28 (a) Express $\sqrt{675}$ in the form $n\sqrt{27}$ where n is a positive integer.

.....

(1)

(b) Show that $\dfrac{5-\sqrt{2}}{\sqrt{2}-1}$ can be written in the form $a + b\sqrt{2}$ where a and b are integers.

(3)

(Total for question = 4 marks)

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29 Solve $\sqrt{3}(x - 2\sqrt{3}) = x + 2\sqrt{3}$

Give your answer in the form $a + b\sqrt{3}$ where a and b are integers.
Show your working clearly.

$x =$ .....

(Total for question = 4 marks)

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30 Express $\dfrac{3 + \sqrt{8}}{\left(\sqrt{2} - 1\right)^2}$ in the form $p + \sqrt{q}$ where p and q are integers.

Show each stage of your working clearly.

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CHALLENGE QUESTIONS

31 The diagram shows a cuboid with a square cross section.

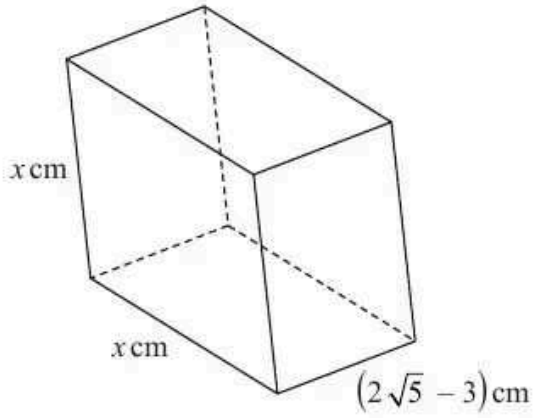


Diagram **NOT** accurately drawn

The volume of the cuboid is $(13 + 6\sqrt{5})\text{cm}^3$

Without using a calculator, find the value of x
Give your answer in the form $a + \sqrt{b}$ where a and b are integers.
Show your working clearly.

$x =$ .....

(Total for Question 23 is 4 marks)

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32 The area of a rectangle is $18\,\text{cm}^2$

The length of the rectangle is $\left(\sqrt{7}+1\right)$ cm.

Without using a calculator and showing each stage of your working,

find the width of the rectangle.
Give your answer in the form $a\sqrt{b}+c$ where a, b and c are integers.

..... cm

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33 Simplify $\dfrac{30\times 25^{2x+7}}{\sqrt{180}\times\left(\sqrt{5}\right)^{4x+9}}$

Give your answer in the form 5^w where w is an expression in terms of x
Show each stage of your working clearly.

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34 Given that $8\sqrt{m} + \sqrt{49m} - \sqrt{9m} = k\sqrt{m}$

where k is an integer and m is a prime number,

(a) work out the value of k

$k =$ .....

(1)

Show that $\dfrac{5 - \sqrt{18}}{1 - \sqrt{2}}$ can be written in the form $a + b\sqrt{2}$

where a and b are integers.
Show each stage of your working clearly.

(3)

(Total for Question 17 is 4 marks)

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35 (a) Show that $\left(6+2\sqrt{12}\right)^2 = 12\left(7+4\sqrt{3}\right)$

Show each stage of your working.

(3)

(b) Simplify fully $\left(\frac{27a^{12}}{t^{15}}\right)^{-\frac{2}{3}}$

.....

(3)

(Total for question = 6 marks)

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TOTAL FOR PAPER IS 108 MARKS

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